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Generalized Linear Models

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Generalized Linear Models

Thanks to Dr. Mu He for teaching me GLM. I keep coming back to projection geometry, likelihood, exponential families and what the model actually means.

MTH303: Linear Statistical Models

I especially like how the history and intuition behind linear models connect to likelihood and modern statistical thinking. This is why I want to understand the proofs, not just the formulas.

⟵ **Proof idea** Check the initial and boundary conditions after finding a general solution.

The notes below bring together the course's regression, inference, categorical-predictor, likelihood, and generalized linear model ideas. Throughout, n is the number of observations; p counts slopes, so a design matrix with an intercept has $p + 1$ columns.

From a mean to a regression model

⟵ **My thought** Decomposing a complicated signal into simpler modes makes dynamics easier to understand.

A sample mean predicts every observation in the same way. It is useful when observations can reasonably be treated as exchangeable around one common mean. If a measured covariate explains systematic differences, a conditional mean is more informative: $m(x) = E(Y \mid X = x)$. In simple linear regression (SLR), we approximate this mean by $\beta_0 + \beta_1 x$. The line is a model for the *mean*, while the random error describes variation around it:

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad E(\varepsilon_i \mid X) = 0, \qquad \mathrm{Var}(\varepsilon_i \mid X) = \sigma^2.$$

For the classical normal theory used below, condition additionally on the design and assume $\varepsilon \sim N_n(0, \sigma^2 I_n)$. Least squares itself does not require normality; the exact small-sample t and F distributions do. A slope is the change in the modeled mean per unit change in x. With several predictors, its interpretation is conditional on the other predictors being held fixed. An intercept describes the modeled mean when the included predictors are zero, which need not be a scientifically meaningful setting.

Least squares, fitted values, and residuals

⟵ **Motivation** Why minimize distance? Orthogonality turns a nonlinear-looking optimization into normal equations.

The least-squares line minimizes $\mathrm{RSS}(b_0, b_1) = \sum_i (y_i - b_0 - b_1 x_i)^2$. Writing $S_{xx} = \sum_i (x_i - \bar{x})^2$ and $S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y})$ gives

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}, \qquad \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i, \qquad e_i = y_i - \hat{y}_i.$$

The fitted line passes through $(\bar{x}, \bar{y})$. Residuals are observed differences from that fitted line; they are not the unobserved errors ε_i. With an intercept, the normal equations imply $\sum_i e_i = 0$ and $\sum_i x_i e_i = 0$. Geometrically, $\hat{y}$ is the orthogonal projection of the response vector y onto the column space of the design matrix, and e is perpendicular to that space. This geometry explains the normal equations without claiming that the realized y must lie on the fitted line.

Variation and the ANOVA comparison

⟵ **Proof idea** Identify the exact statement before proving it.

With an intercept, $y_i - \bar{y} = (\hat{y}_i - \bar{y}) + e_i$, and the two terms are orthogonal. Therefore

$$\underbrace{\sum_i (y_i - \bar{y})^2}_{\text{TSS}} = \underbrace{\sum_i (\hat{y}_i - \bar{y})^2}_{\text{RegSS}} + \underbrace{\sum_i e_i^2}_{\text{RSS}}, \qquad R^2 = 1 - \frac{\text{RSS}}{\text{TSS}}.$$

TSS has $n-1$ degrees of freedom. For SLR, RegSS has one and RSS has $n-2$; for full-rank multiple regression with p slopes, these become p and $n-p-1$. The residual mean square $s^2 = \text{RSS}/(n-p-1)$ estimates σ^2 under the model. The global statistic

$$F = \frac{\text{RegSS}/p}{\text{RSS}/(n-p-1)}$$

tests whether all slopes are zero. A large F is evidence that the predictors explain variation beyond an intercept-only model. R^2 describes the fraction of sample variation explained relative to that baseline; it does not by itself establish causality or whether the assumptions are adequate. In SLR, the two-sided slope t test and the global F test agree because $F = t^2$.

Uncertainty for a coefficient and for a future response

⟵ **My thought** The theorem is easier to remember when its hypothesis feels necessary.

Under the normal SLR assumptions, $\hat{\beta}_1 \sim N(\beta_1, \sigma^2/S_{xx})$, so a t statistic replaces σ with s. A two-sided confidence interval is $\hat{\beta}_1 \pm t_{n-2,1-\alpha/2} s/\sqrt{S_{xx}}$. The slope test asks about the conditional mean; predicting a new individual response also requires accounting for a new error term.

At a chosen x_*, $\hat{m}(x_*) = \hat{\beta}_0 + \hat{\beta}_1 x_*$. Its estimated standard error is $s\sqrt{1/n + (x_* - \bar{x})^2/S_{xx}}$, whereas the predictive standard error for a future Y_* is

$$s\sqrt{1 + \frac{1}{n} + \frac{(x_* - \bar{x})^2}{S_{xx}}}.$$

Thus a prediction interval uses $\hat{m}(x_*) \pm t_{n-2,1-\alpha/2}$ times the second standard error. It is wider than the corresponding interval for the mean because it includes the future observation's intrinsic variation.

Multiple regression as a projection

⟵ **Motivation** Why minimize distance? Orthogonality turns a nonlinear-looking optimization into normal equations.

For p slopes, write $y = X\beta + \varepsilon$, where X is $n \times (p+1)$ and includes a column of ones. If X has full column rank, least squares gives

$$\hat{\beta} = (X^T X)^{-1} X^T y, \qquad H = X(X^T X)^{-1} X^T, \qquad \hat{y} = Hy, \qquad e = (I - H)y.$$

H is the *hat matrix*: $H^T = H$ and $H^2 = H$. It projects onto $\text{col}(X)$, while $M = I - H$ projects onto its orthogonal complement. Hence $X^T e = 0$, $\hat{y}^T e = 0$, and a full-rank model leaves $n - p - 1$ residual degrees of freedom. An interaction term, such as $x_1 x_2$, allows the slope of one continuous predictor to depend on the other; with a binary z, $\beta_0 + \beta_1 x + \beta_2 z + \beta_3 xz$ gives an intercept change β_2 and a slope change β_3 between groups.

Under $\varepsilon \sim N_n(0, \sigma^2 I)$ conditional on X,

$$\hat{\beta} \mid X \sim N_{p+1}(\beta, \sigma^2 (X^T X)^{-1}), \quad e \mid X \sim N_n(0, \sigma^2 M), \quad \frac{\text{RSS}}{\sigma^2} = \frac{e^T e}{\sigma^2} \sim \chi^2_{n-p-1}.$$

The last result follows because M is symmetric and idempotent with rank $n - p - 1$: in a suitable orthonormal basis, its quadratic form is the sum of that many squared independent standard normals. Moreover,

$$\operatorname{Cov}(e, \hat{\beta} \mid X) = \sigma^2 M X (X^T X)^{-1} = 0.$$

Both are linear transformations of a jointly normal vector, so zero covariance makes e and $\hat{\beta}$ independent. The same is true of e and $\hat{y}$. Under the intercept-only null, $\mathrm{RegSS}/\sigma^2 \sim \chi^2_p$, independently of RSS/σ^2, which leads to the global F test above. These exact independence statements use the normal-error model; orthogonality of fitted values and residuals alone is a geometric fact.

Single and grouped questions

⟵ **Proof idea** Identify the exact statement before proving it.

A single coefficient can be tested with $t = (\hat{\beta}_j - \beta_{j,0})/[s\sqrt{(X^T X)^{-1}_{jj}}]$, using $n-p-1$ degrees of freedom. For a group of q coefficients, fit a *reduced* model under the restriction and a *full* model containing the additional terms. Then

$$F = \frac{(\mathrm{RSS}_R - \mathrm{RSS}_F)/q}{\mathrm{RSS}_F/(n-p-1)} \sim F_{q,n-p-1} \quad \text{under } H_0.$$

This partial F test asks whether the group improves fit after the other variables have been included. It is more appropriate for a joint hypothesis than choosing among several individual t tests. The models must be nested, and both RSS values must be computed from the same responses.

Categorical predictors: a change of coordinates

⟵ **My thought** The theorem is easier to remember when its hypothesis feels necessary.

A categorical predictor with r levels needs $r-1$ independent columns when the model has an intercept. Including all r indicators as well as the intercept makes X rank deficient, because the indicators sum to the intercept column. With treatment coding, the omitted level is a reference group, and each displayed coefficient is a difference from its mean. With effect coding, coefficients describe deviations under the chosen sum-to-zero constraint. The numerical coefficients and their interpretation change with the coding, while the fitted group means need not.

The tutorial's education example has three levels: no high school, completed high school, and post-secondary. Its effect-coded rows for $(1, E, F)$ are $(1,1,0)$, $(1,0,1)$, and $(1,-1,-1)$. Treatment coding with post-secondary as reference instead uses $(1, D_N, D_C)$ with rows $(1,1,0)$, $(1,0,1)$, and $(1,0,0)$. For the same three observations these design matrices satisfy $X_{\text{effect}} = X_{\text{dummy}} T$, where

$$T = \begin{pmatrix} 1 & -1 & -1 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}, \qquad \det(T) = 3 \neq 0.$$

Thus $\operatorname{col}(X_{\text{effect}}) = \operatorname{col}(X_{\text{dummy}})$. More generally, any invertible column change $X_2 = X_1 T$ gives $\beta_2 = T^{-1}\beta_1$, the same fitted values, and the same projection matrix $H_2 = H_1$. This is *matrix reparameterization*: an individual coefficient may look different, but a properly translated contrast and its inference describe the same group comparison. For two categorical factors, both parameterizations must also represent the same intended main effects and interactions before their fits can be compared.

In the tutorial's additive wage example, education uses (E, F) and number of children uses (G, H), each with three levels. The mean for post-secondary education and more than two children is $a - b_1 - b_2 - c_1 - c_2$ under the stated effect coding. Its contrast with the balanced average of all nine education-by-children cell means is $-b_1 - b_2 - c_1 - c_2$. A sample-weighted average is a different target when group sizes are unequal; the reference average must be specified before interpreting a contrast.

Likelihood and maximum likelihood estimation

⟵ **Motivation** Inference uses data to distinguish plausible parameters; assumptions still matter.

For independent observations with density or mass $f(y_i;\theta)$, the likelihood is $L(\theta; y) = \prod_i f(y_i;\theta)$ and the log-likelihood is $\ell(\theta; y) = \sum_i \log f(y_i;\theta)$. The data are fixed in this optimization; θ varies. Maximizing L or ℓ gives the same estimate. A zero derivative identifies a candidate stationary point, not automatically a global maximum or a boundary solution.

For Bernoulli responses, the MLE of their common success probability is the sample proportion. For independent exponential observations parameterized by rate λ, $\widehat{\lambda} = 1/\bar{y}$. Under a normal model, the likelihood estimators are $\hat{\mu} = \bar{y}$ and $\hat{\sigma}^2_{\text{MLE}} = n^{-1}\sum_i (y_i - \bar{y})^2$; the unbiased sample variance uses $n-1$ instead. This distinction also appears in normal regression: the MLE of σ^2 divides RSS by n, while the usual residual mean square divides by the residual degrees of freedom.

Generalized linear models

⟵ **Proof idea**
Check the initial and boundary conditions after finding a general solution.

A GLM keeps a linear *predictor* $\eta_i = x_i^T\beta$ while allowing the response distribution and mean-to-predictor relationship to change. It has three parts:

1. A random component: conditionally independent Y_i from a specified exponential dispersion family.
2. A systematic component: $\eta_i = x_i^T\beta$.
3. A link: $g(\mu_i) = \eta_i$, where $\mu_i = E(Y_i \mid x_i)$.

A positive-valued outcome illustrates why this matters. Fitting a straight line on the original scale can predict negative means. Regressing $\log Y$ changes the modeled response and targets $E(\log Y \mid x)$, which is not generally $\log E(Y \mid x)$. A GLM with a log link instead specifies $\log E(Y \mid x) = x^T\beta$ directly, with an appropriate response family and variance. This distinction is important even when both approaches can produce visually straight plots after a transformation.

Exponential family, mean, and variance

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

Write an exponential dispersion family as

$$f(y;\theta,\phi) = \exp\left\{\frac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)\right\}.$$

Here θ is the natural parameter and ϕ controls dispersion. Differentiating the normalized density and using the expected score equal to zero gives $\mu = b'(\theta)$. Differentiating again gives $\operatorname{Var}(Y) = a(\phi)b''(\theta)$. The *variance function* is $V(\mu) = b''(\theta(\mu))$, so $\operatorname{Var}(Y) = a(\phi)V(\mu)$. The link is *canonical* when $g(\mu) = \theta(\mu)$.

Response family	Mean μ	Variance function $V(\mu)$	Common link
Gaussian	μ	1	Identity
Bernoulli	μ	$\mu(1-\mu)$	Logit
Poisson	μ	μ	Log
Gamma	μ	μ^2	Inverse (up to parameterization)
Inverse Gaussian	μ	μ^3	Inverse-squared (up to scale)

The mean-variance relationship is part of the model, not merely a nuisance. For Bernoulli outcomes, the variance depends on the mean even when observations are otherwise independent. The canonical parameterization is not the only possible parameterization, and a valid GLM need not use its canonical link.

Links and interpretations

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

With a Gaussian response and identity link, $\mu_i = x_i^T\beta$ recovers ordinary normal linear regression. For Poisson counts, $\log \mu_i = x_i^T\beta$ guarantees $\mu_i > 0$ and gives $\mu_i = \exp(x_i^T\beta)$; a one-unit change in a predictor multiplies the modeled mean by e^{β_j} when other predictors are held fixed. For binary outcomes, the logistic model uses

$$\log\frac{\pi_i}{1-\pi_i} = x_i^T\beta, \qquad \pi_i = \frac{e^{x_i^T\beta}}{1+e^{x_i^T\beta}},$$

so e^{β_j} is a conditional odds ratio for a one-unit change. The probit alternative uses $\Phi^{-1}(\pi_i) = x_i^T\beta$ and therefore $\pi_i = \Phi(x_i^T\beta)$. Both keep fitted probabilities between zero and one; their coefficient scales differ.

Score equations and a working response

⟵ **Proof idea** Identify the exact statement before proving it.

Let $g(\mu_i) = \eta_i = x_i^T\beta$ and $\theta_i = \theta(\mu_i)$. Applying the chain rule to the log-likelihood gives the score component

$$U_j(\beta) = \sum_{i=1}^{n} \frac{(y_i - \mu_i)x_{ij}}{a(\phi)V(\mu_i)g'(\mu_i)}.$$

Define $w_i = 1/[V(\mu_i)\{g'(\mu_i)\}^2]$ and the local working response $z_i = \eta_i + g'(\mu_i)(y_i - \mu_i)$. Then

$$U(\beta) = \frac{1}{a(\phi)} X^T W(z - X\beta), \qquad W = \text{diag}(w_1, \dots, w_n).$$

These weights and responses depend on the current β; solving $X^TWX\beta = X^TWz$ therefore means updating them iteratively, rather than performing one ordinary least-squares fit. Locally, z_i linearizes the link around the current fitted mean, and w_i reflects its variance. For normal errors with an identity link, $w_i = 1$ and $z_i = y_i$, reproducing $X^T(y - X\beta) = 0$. For a Poisson log-link model, $V(\mu_i) = \mu_i$, $g'(\mu_i) = 1/\mu_i$, hence $w_i = \mu_i$ and $z_i = \eta_i + (y_i - \mu_i)/\mu_i$; the score reduces to $X^T(y - \mu) = 0$.

The practical lesson across all these models is to choose the response distribution, link, predictor coding, and inferential target together. A familiar coefficient or small p-value is only useful after its model and comparison have been made explicit.

Linear Regression

Partial modeling-P(x|y)

⟵ **My thought** A linear predictor connects geometry to prediction, but the link function changes the response scale.

For many points, there are many possible regression functions f (nonparametric) so it is useful to restrict to simple functions described by a few parameters.

Though true regression functions are never linear, we assume the dependence of Y on $X_1 \dots X_n$ is linear.

regression function: x—-> f(x) :

$$E[Y|X = x] = \int yh(y|x)dy$$

(ps: the conditional median:m(x)

$$\int_{-\infty}^{m(x)} h(y|x)dy = 1/2$$

)

Simple linear regression

- **OLS(Ordinary Least-Squares)**: Historically, the OLS method was introduced before the systematic understanding of conditional expectation. The OLS method was proposed by French mathematician Adrien-Marie Legendre in 1805 and later developed by Carl Friedrich Gauss in 1809. The OLS method was created to find the regression coefficients that best fit the data. An interesting geometry explaination in Linear Algebra's projection matrix of OLS in multiple(simple) linear regression:

- **Conditional Expectation**: After the OLS method was established, it was recognized that the regression equation obtained through OLS is actually the expected value of (Y) given (X), i.e., the conditional expectation (E(Y|X=x)). Under the classical linear regression assumptions, the OLS estimate is an unbiased estimate of the conditional expectation.

Specifically, for a simple linear regression model (

$$Y = \beta_0 + \beta_1 X + \epsilon$$

):

- **OLS Estimate**: Through the OLS method, we obtain the estimated regression coefficients ($\hat{\beta}_0$) and ($\hat{\beta}_1$), thus writing the regression equation (

  $$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$$

). (based on the smallest RSS)
- **Conditional Expectation**: According to the model assumption, the conditional expectation (E(Y|X=x)) should be ($\beta_0 + \beta_1 x$)(which is called the real regression line(not the true line) and could not be seen in practice). The regression equation obtained through OLS, ($\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$), is an estimate of (E(Y|X=x)).

In summary, OLS(similar to MLE in SLR) is the best way to estimate the true model, which generates the unbiased estimators of the real regression line that based on $E[y_i|x]$.

Another explaination-analogy of population mean in sampling distribution:

essence of SLR relavant to ρ (correlation coefficient)

$$\hat{y} - \bar{y}/sd(\hat{y}) = \rho(\hat{x} - \bar{x}/sd(\hat{x}))$$

Proof using covariance when it is normal distribution(Teacher:shisong mao)

Hypothesis test in (simple/multiple-the same philosophy) Linear Regression

t-statistics for estimators z—-t—-F

slope H_0:β_1=0 ###### t for testing significance of regression Because $y_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$~NID($\beta_0 + \beta_1 x_i, \sigma^2$), β_1~N(β_1,$\sigma^2\Sigma(x_i - \bar{x})^2$ so if σ is known,

$$\hat{\beta}_1 / \sqrt{\sigma^2 / \Sigma(x_i - \bar{x})^2}$$

but if σ is unknown,

$$t_0 = \hat{\beta}_1 / se(\hat{\beta}_1)$$

, which just substitute σ's unbiased estimation($\Sigma(y_i - \hat{y})^2/n - 2$, which is RSS)for σ.

F of analysis of variance for testing the significance of regression(2 ways to understand F test—model interpretation and hypothesis test with noncentralirity parameter(0 or not 0)

$$\frac{\Sigma(y_i - \hat{y})^2}{\sigma^2 p} / \frac{\Sigma(\hat{y} - \bar{y})^2}{\sigma^2(n - p - 1)} \sim F$$

The F-statistic essentially compares the amount of systematic variance explained by the model to the amount of unsystematic variance (error) and helps determine the overall significance of the model. This is why the equation can express the model's explanatory power andrelationship with hypothesis testing.

If H_0 is true, F is a centrality parameter. If we reject H_0, F follows 1 and n-2 degrees of freedom for simple linear regression and a noncentrality parameter is

$$\lambda = \beta_1^2 \frac{\Sigma(x_i - \bar{x})^2}{\sigma^2}$$

⟵ **Motivation**
From a probabilistic model to a usable numerical quantity: what is observed, what is random?

and λ is not = 0. Also, F represents the ability of model's explanation, so if F> 1, or > the certain threshold, the model is good.

As for one variable's test, t is more adaptable because of its one-sided alternative hypotheses although it is equivalent to the F test for the simple linear regression.

the statistical properties of the maximum-likelihood estimators(2.10)

Logistic regression

Instead of linear OLS regression we use something called logistic regression, which is probably more widely used (especially if include variants) than linear OLS. ## why not SLR? (1)SLR may predict values that are below zero or above 1 (2)For SLR we assumed that the population distribution was normally distributed around the mean, for each value of the X variable. That's not going to be the case if we've got a binary response.

⟵ **Proof idea**
Distinguish a model for the conditional mean from the noise model.

Test the model

ML-LRT-compare the maximum of the likelihood function for each model

cross validation ###r example of logistic regression with variable selection and model selection ####Check the goodness of fit—-Hosmer-Lemeshow GOF test is the most widely used for logistic regression model. ### r example

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

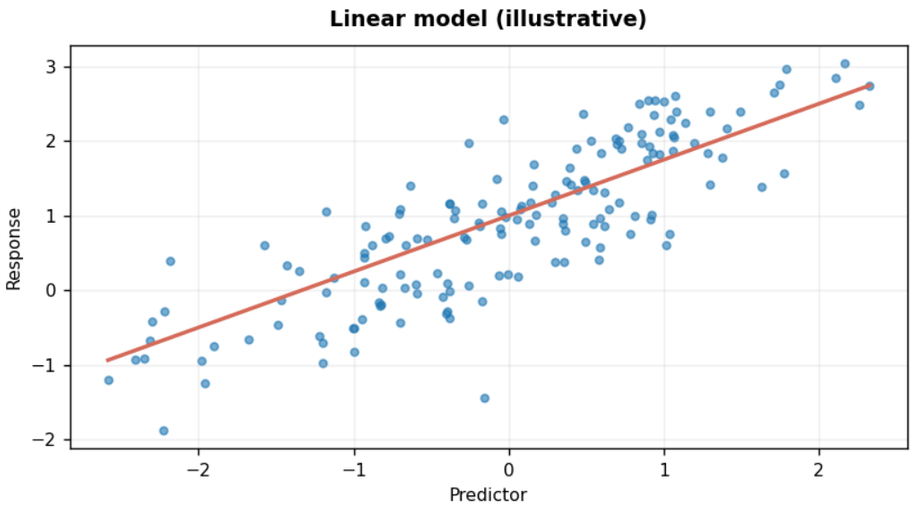


Figure 1: Output from code example 1. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ISLR2, ISLR; functions: plot, glm, predict.

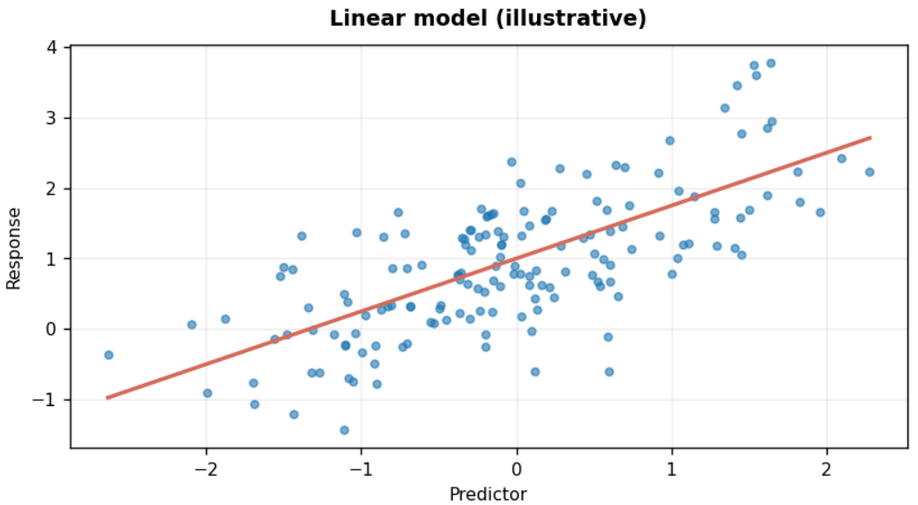


Figure 2: Output from code example 1. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ISLR2, ISLR; functions: plot, glm, predict.

MTH303 additions —how I connect the ideas

Thanks to Dr. Mu He for teaching me GLM.

My favourite thing about this subject is that the ideas do not stay in separate boxes. Least squares introduces projections; projections explain degrees of freedom; likelihood suggests how to handle new response distributions; and the same linear algebra returns when I fit a GLM numerically. The following are **my own synthesis notes**, based on the eight further-reading resources linked from the MTH303 teaching page. I have reconstructed the arguments, examples and derivations in my own order and words rather than reproducing the online handouts. I have omitted full code listings as in my other notes.

Road map: what I want to understand

1. Why a projection gives the normal equations, ANOVA, t and F tests.
2. Why different model-selection penalties answer different questions.
3. Why the exponential family gives the GLM its mean-variance relationship.
4. Why a canonical link can simplify the likelihood score.
5. How Newton's method, Fisher scoring and IRLS relate.
6. What deviance measures, including its difficult boundary cases.
7. How I recognize a GLM family, its link, weights and residuals.
8. Why cross-validation is a different exercise from fitting a model once.

I. Chapter 3: multiple linear regression and the hat matrix

Projection is the mathematical starting point

Let $X \in \mathbb{R}^{n\times q}$ have full column rank $q = p + 1$. In the Gaussian linear model,

$$Y = X\beta + \varepsilon, \qquad E(\varepsilon \mid X) = 0, \qquad \operatorname{Var}(\varepsilon \mid X) = \sigma^2 I.$$

I try to remember the objective function by thinking of the residual vector itself, rather than an arbitrary formula:

$$S(b) = (y - Xb)^\top (y - Xb) = y^\top y - 2b^\top X^\top y + b^\top X^\top X b.$$

Differentiating with respect to b yields

$$\nabla S(b) = -2X^\top y + 2X^\top X b.$$

At a minimum the derivative is zero, hence the normal equations $X^\top X\hat\beta = X^\top y$. Full rank implies

$$\boxed{\hat\beta = (X^\top X)^{-1} X^\top y}.$$

The matrix $H = X(X^\top X)^{-1}X^\top$ projects onto the column space $\mathcal{C}(X)$. Because $H^2 = H$ and $H^\top = H$, I can use orthogonality instead of doing every sum of squares by hand:

$$\hat y = Hy, \qquad e = (I - H)y, \qquad X^\top e = 0, \qquad \hat y^\top e = 0.$$

> **My margin thought.** The hat matrix is a geometric object, not just a device for computing fitted values. The whole regression story becomes easier when I separate the fitted direction from the unexplained direction.

Degrees of freedom and the unbiased variance estimator

Since H is an orthogonal projection of rank q, $I - H$ has rank $n - q$. Under independent Gaussian errors,

$$\frac{\mathrm{RSS}}{\sigma^2} = \frac{\varepsilon^\top (I - H)\varepsilon}{\sigma^2} \sim \chi^2_{n-q}.$$

MY SIDE NOTE

Regression geometry

I understand regression by projecting y onto the columns of X. The residual is the orthogonal component.

Question I would ask: what assumption, derivative or geometric identity justifies this step?

Therefore $E(\mathrm{RSS}) = (n-q)\sigma^2$. This is why I divide the residual sum of squares by $n-q$ and not by n:

$$\boxed{s^2 = \frac{\mathrm{RSS}}{n-p-1}}, \qquad E(s^2) = \sigma^2.$$

The identity $\mathrm{tr}(H) = q$ is also the starting point of leverage. For observation i, h_{ii} measures how much the fitted value at that observation depends on its own response. The average leverage is q/n, because $\sum_i h_{ii} = q$.

ANOVA and exact distributions

With an intercept, let $H_0 = \mathbf{1}\mathbf{1}^\top/n$ be projection onto constant vectors. The centred response decomposes as

$$y - \bar{y}\mathbf{1} = (Hy - \bar{y}\mathbf{1}) + (y - Hy).$$

The two terms are orthogonal, so

$$\mathrm{TSS} = \mathrm{RegSS} + \mathrm{RSS}\,.$$

In a Gaussian model under the global null that all slopes vanish, the projection subspaces associated with regression and residual variation are orthogonal. Thus

$$\frac{\mathrm{RegSS}}{\sigma^2} \sim \chi^2_p, \qquad \frac{\mathrm{RSS}}{\sigma^2} \sim \chi^2_{n-p-1},$$

independently, and

$$F = \frac{\mathrm{RegSS}\,/p}{\mathrm{RSS}\,/(n-p-1)} \sim F_{p,n-p-1}.$$

I need to check the conditions here: the projection identities are algebraic, but the exact finite-sample χ^2, t and F laws need Gaussian assumptions (or a separate justification).

Partial F, single-coefficient t, and the identity $t^2 = F$

Suppose the reduced model has q_R free coefficients and the full model has q_F, where $r = q_F - q_R$ restrictions are being tested. Then

$$F_{\text{partial}} = \frac{(\mathrm{RSS}_R - \mathrm{RSS}_F)/r}{\mathrm{RSS}_F\,/(n-q_F)}.$$

In the special case of a single restriction $r = 1$, the reduction in RSS is exactly the squared coefficient estimate divided by the relevant diagonal of $(X^\top X)^{-1}$. Hence the partial F statistic equals the square of the corresponding t statistic. I find this much more convincing than memorizing $t^2 = F$ as an isolated fact.

Studentized residuals and leverage

The diagonal of the residual covariance is $\mathrm{Var}(e_i \mid X) = \sigma^2(1-h_{ii})$. This motivates standardized and studentized residuals: a raw residual of size 1 means different things at different leverage values. For influence, I look jointly at residual size, leverage and the change in the fit caused by deleting the point. A high-leverage observation does not automatically have a large residual, and a large residual does not automatically have high leverage.

Original source for these chapter connections: MTH303 Chapter 3 teaching notes. The original resource also includes a detailed eight-observation worked example and a hand-versus-R comparison. For a reproducible check in my own R session I would use `lm()`, `model.matrix()`, `crossprod()`, `solve()`, `hatvalues()`, `anova()`, and `rstandard()`.

MY SIDE NOTE

Least squares & tests

I want to separate algebraic ANOVA identities from the Gaussian assumptions behind F and t distributions.

Question I would ask: what assumption, derivative or geometric identity justifies this step?

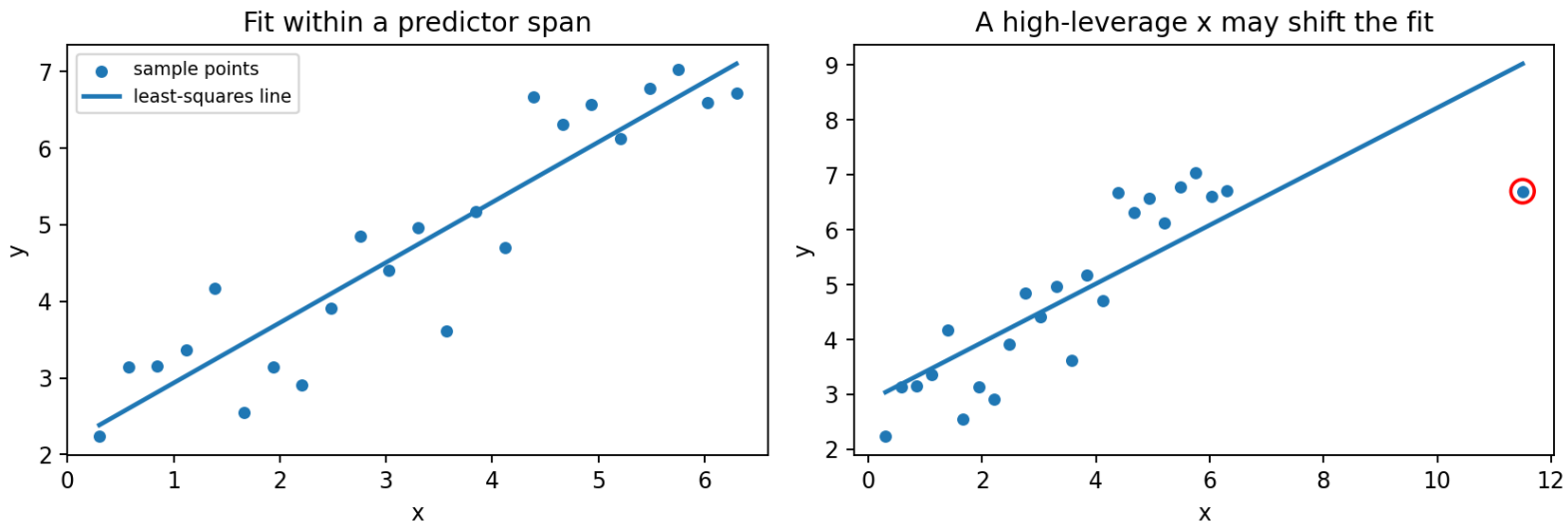


Figure 1: My reconstruction: projection geometry and leverage in two regressions. R equivalents: `lm()`, `hatvalues()`, `plot()` or `ggplot2::geom_point()` plus `geom_smooth()`.

II. Why model-selection criteria have different penalties

One more variable always helps the training RSS

Least squares chooses the coefficients after seeing the data. Enlarging the column space gives the optimizer more directions, so RSS cannot increase. That does not mean the new variable really helps prediction. I want to distinguish a statement about the **training projection** from one about the **distribution of future observations**.

Adjusted R^2 is a degrees-of-freedom correction

For p slopes and an intercept,

$$R^2 = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}}, \qquad \boxed{R^2_{\mathrm{adj}} = 1 - \frac{\mathrm{RSS}\,/(n-p-1)}{\mathrm{TSS}\,/(n-1)}}.$$

I read this as a ratio of two variance estimates. The denominator loses 1 degree of freedom because I estimated a mean; the numerator loses $p+1$ because I fitted the regression coefficients. This is the same degrees-of-freedom logic that appeared in the unbiased variance estimator.

AIC asks about predictive information loss

For k estimated parameters and maximized log-likelihood $\ell(\hat{\theta})$,

$$\boxed{\mathrm{AIC} = -2\ell(\hat{\theta}) + 2k}.$$

The correction $2k$ compensates approximately for the optimistic evaluation of fitted likelihood on the same observations. In regular large-sample models, the argument can be connected to expected out-of-sample Kullback–Leibler information loss. I do not interpret the numerical AIC itself as an absolute measure of goodness: comparisons require a common response, consistent likelihood conventions and the same data.

BIC asks a different question

$$\boxed{\mathrm{BIC} = -2\ell(\hat{\theta}) + k\log n}.$$

MY SIDE NOTE

Model choice

I read AIC, BIC and adjusted R-squared as different answers to different questions.

Question I would ask: what assumption, derivative or geometric identity justifies this step?

The $k \log n$ penalty arises from a large-sample approximation to integrated likelihood (under suitable priors and regularity conditions). Because its target and assumptions differ from AIC, I should not say BIC is just a "stricter AIC" without qualification. It often prefers smaller models for large n, but prediction and evidence for a model are different aims.

For $n > e^2$ the BIC penalty per parameter exceeds AIC' s; for smaller samples and different effective parameter-counting rules I need to inspect the conventions. I also need to include variance and dispersion parameters consistently when comparing full likelihood-based models.

An example of conflicting decisions

Suppose adding one coefficient raises maximized log likelihood by 3. Then the difference in twice negative log likelihood is -6. The AIC change is $-6 + 2 = -4$, favouring the larger model. At $n = 1000$, the BIC change is approximately $-6+\log(1000) = +0.91$, favouring the smaller model. Neither criterion has contradicted arithmetic: they have different inferential targets.

What I would write next to the model-selection table: adjusted R^2 compares explained variance adjusted for df; AIC approximates a prediction-oriented information criterion; BIC approximates an evidence-oriented criterion. All depend on the model class and assumptions.

Based on: Origins of adjusted R², AIC and BIC, including its historical context, shrinkage motivation, and derivations.

III. Chapter 4: why a GLM needs the exponential family

Why normal linear regression is not always sensible

If the response is a count, a real-valued linear prediction may become negative. If it is a probability, an identity-link fitted value may fall outside $[0, 1]$. If the variance grows with the mean, the constant variance assumption is also unsuitable. I now think of a GLM as a **two-part modeling choice**: a response distribution and a link from its conditional mean to a linear predictor.

$$\eta_i = x_i^\top \beta, \qquad g(\mu_i) = \eta_i, \qquad \mu_i = E(Y_i \mid x_i).$$

These equations do not by themselves determine the variance: it comes from the family.

My favourite derivation: mean and variance from one formula

Start with the exponential dispersion family

$$f(y; \theta, \phi) = \exp\left\{ \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi) \right\}.$$

For a fixed ϕ, differentiating the log likelihood with respect to θ gives

$$s_\theta(y) = \frac{y - b'(\theta)}{a(\phi)}, \qquad \frac{\partial^2 \ell}{\partial \theta^2} = -\frac{b''(\theta)}{a(\phi)}.$$

Because the expected score is zero under regularity conditions,

$$\boxed{\mu = E(Y) = b'(\theta)}.$$

Equating the score variance with negative expected curvature produces

$$\boxed{\mathrm{Var}(Y) = a(\phi) b''(\theta) = a(\phi) V(\mu)}.$$

I like this proof because it tells me why the mean and variance functions are linked rather than arbitrary unrelated assumptions. The dispersion term determines scale, while $V(\mu)$ determines the shape of the mean–variance relationship.

MY SIDE NOTE

Exponential families

I derive the mean and variance from b(theta) instead of learning the family table by heart.

Question I would ask: what assumption, derivative or geometric identity justifies this step?

Distribution examples I should be able to derive

For Bernoulli, $\theta = \log[\pi/(1-\pi)]$ and $b(\theta) = \log(1+e^{\theta})$. Therefore $b'(\theta) = \pi$ and $b''(\theta) = \pi(1-\pi)$. For Poisson, $\theta = \log \mu$, $b(\theta) = e^{\theta}$ and $V(\mu) = \mu$. For a normal observation with known variance, the natural parameter is the mean and the variance function is constant. For a Gamma mean parameterization, the variance function is proportional to μ^2; for inverse Gaussian it is proportional to μ^3.

The crucial question I ask myself is: **what is a property of the distribution and what is a choice I impose with the link?** I should never treat the log link as an intrinsic part of Poisson probability, or the logit link as an intrinsic part of Bernoulli probability.

Based on: Chapter 4 GLM proofs. Its exponential-family examples and derivation checks motivated the order I use here.

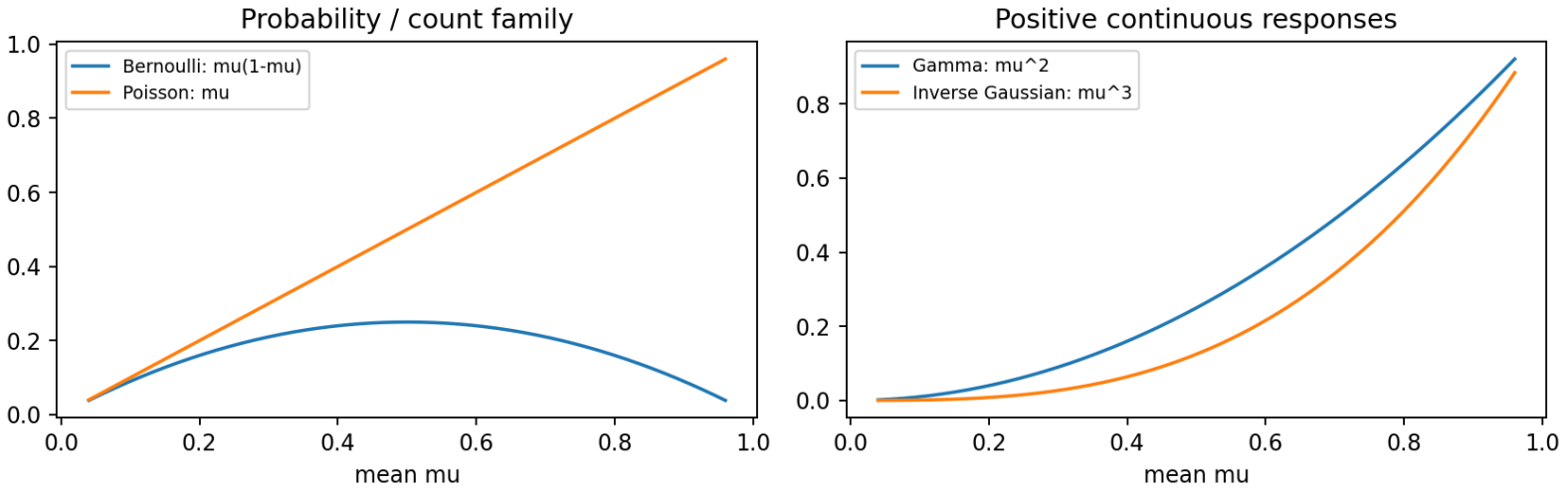


Figure 2: My reconstruction: mean-variance functions for Bernoulli, Poisson, Gamma and inverse Gaussian models. R equivalents: base `curve()`, `plot()`, and `lines()`, or `ggplot2::geom_line()`.

IV. The parameter chain: canonical and non-canonical links

Keep three quantities separate

The most important diagram in my GLM notes is

$$\boxed{\beta} \xrightarrow{x_i^\top \beta} \boxed{\eta_i} \xrightarrow{g^{-1}} \boxed{\mu_i} \xrightarrow{(b')^{-1}} \boxed{\theta_i}.$$

The linear predictor η, conditional mean μ, and exponential-family natural parameter θ are normally three different quantities. Here $h(\eta) = (b')^{-1}(g^{-1}(\eta))$ maps from η to θ. The data do not give me three independent parameter vectors: choosing the regression coefficients, family and link fixes this chain.

Why the canonical link is special

The canonical link is chosen as $g(\mu) = (b')^{-1}(\mu)$, which makes $\eta = \theta$. This simplifies score equations; it does not mean non-canonical links are invalid.

For Bernoulli, with $\mu = 0.8$, the natural parameter and logit predictor are $\log 4 \approx 1.3863$. Under probit, the predictor is $\Phi^{-1}(0.8) \approx 0.8416$, although the conditional mean is unchanged. The number 0.8 is not the coefficient or the linear predictor; it is the mean or success probability.

Derivatives are a chain, not a memorized miracle

I write the full derivative to avoid dropping a factor:

MY SIDE NOTE

Link functions

I draw beta -> eta -> mu -> theta every time I take derivatives.

Question I would ask: what assumption, derivative or geometric identity justifies this step?

$$\frac{\partial \ell}{\partial \beta_j} = \frac{\partial \ell}{\partial \theta}\frac{\partial \theta}{\partial \mu}\frac{\partial \mu}{\partial \eta}\frac{\partial \eta}{\partial \beta_j}.$$

Because $d\mu/d\theta = V(\mu)$ and $d\eta/d\mu = g'(\mu)$,

$$\boxed{\frac{\partial \ell}{\partial \beta_j} = \sum_i \frac{(y_i - \mu_i)x_{ij}}{a(\phi)V(\mu_i)g'(\mu_i)}}.$$

For a canonical link $g'(\mu) = 1/V(\mu)$, the denominator simplifies to $a(\phi)$, giving a score proportional to $X^\top(y - \mu)$. This is why the canonical-link score looks surprisingly similar to the normal equations.

A warning about link domains

A log link guarantees a positive mean. An inverse link also constrains the admissible linear predictor, often requiring $\eta > 0$ or $\eta < 0$ depending on its sign convention. Even a mathematically valid link must be checked for compatibility with the desired fitted means and predictor values.

Based on: Canonical and non-canonical links, including its Bernoulli logit-versus-probit explanation and chain-rule treatment.

V. Chapter 5: MLE, Fisher information, and IRLS

The score equation

With independent observations and fixed dispersion, the log likelihood is

$$\ell(\beta) = \sum_i \left[\frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi)\right].$$

The score follows from the derivative chain in the previous section. Unlike ordinary least squares, setting it to zero usually gives a nonlinear system that needs numerical iteration.

Expected information and weights

Writing $d\mu_i/d\eta_i = 1/g'(\mu_i)$, the Fisher information can be expressed as

$$\mathcal{J}(\beta) = \frac{1}{a(\phi)}X^\top W X, \quad W = \operatorname{diag}\left\{\frac{1}{V(\mu_i)[g'(\mu_i)]^2}\right\}.$$

I find the weight formula especially memorable: **high variance and a steep link both weaken information about the linear predictor**.

Newton–Raphson, Fisher scoring and working least squares

Let $U(\beta) = \partial\ell/\partial\beta$ and $J(\beta) = -\partial^2\ell/\partial\beta\partial\beta^\top$ be observed information. A Newton update for maximizing likelihood is

$$\beta^{(t+1)} = \beta^{(t)} + J(\beta^{(t)})^{-1}U(\beta^{(t)}).$$

Fisher scoring replaces J with $\mathcal{J}$. I can obtain the same scoring step by solving a weighted least-squares problem with working response

$$z_i = \eta_i + (y_i - \mu_i)g'(\mu_i), \qquad \boxed{\beta^{(t+1)} = (X^\top W X)^{-1}X^\top W z}.$$

This is iterative reweighted least squares, or IRLS. It does not mean GLM estimation is ordinary unweighted regression; both W and z change with $\beta^{(t)}$.

MY SIDE NOTE

Link functions

I draw beta -> eta -> mu -> theta every time I take derivatives.

Question I would ask: what assumption, derivative or geometric identity justifies this step?

Why canonical models simplify further

In many canonical-link exponential-family models, the observed and expected information agree in the relevant coefficient curvature, so Newton and Fisher scoring coincide. For a noncanonical link, terms involving response residuals may remain in the observed Hessian and the two updates generally differ.

For Bernoulli-logit models, $W_{ii} = \pi_i(1-\pi_i)$ and $z_i = \eta_i + (y_i - \pi_i)/[\pi_i(1-\pi_i)]$. This is the easiest IRLS identity for me to derive independently.

Things I check before trusting the algorithm

I check that the design matrix has sufficient rank, fitted means remain inside the permissible domain, numerical weights are not singular, and the likelihood or deviance stabilizes. Complete separation in logistic regression can make the finite MLE fail to exist even though iterative estimates appear to grow steadily. For noncanonical links the optimization may need cautious starting values or damping.

Based on: Chapter 5: GLM MLE.

VI. Deviance: measuring discrepancy from a saturated fit

I understand deviance as a likelihood gap

Let $\ell(\hat{\mu})$ be the fitted model log likelihood and ℓ_{sat} the largest log likelihood for a saturated model with one mean per observation. Up to the family-specific dispersion convention, deviance is

$$\boxed{D = 2a(\phi)\{\ell_{\text{sat}} - \ell(\hat{\mu})\}}.$$

For Bernoulli and Poisson, $a(\phi) = 1$, so there is no additional scaling. The saturated model is a benchmark for comparison, not necessarily a useful predictor.

Bernoulli deviance and the boundary limit

For individual Bernoulli observations $y_i \in \{0, 1\}$ with fitted probability $\hat{\pi}_i$,

$$D = 2\sum_i \left[y_i \log \frac{y_i}{\hat{\pi}_i} + (1-y_i) \log \frac{1-y_i}{1-\hat{\pi}_i} \right].$$

We use $0 \log 0 = 0$ by continuity. I cannot substitute the saturated-model natural parameter directly because $\text{logit}(0) = -\infty$ and $\text{logit}(1) = +\infty$. Instead, in $y\theta - \log(1 + e^{\theta})$ I take the combined limit: for $y = 1$, the expression approaches zero as $\theta \to +\infty$; for $y = 0$ it approaches zero as $\theta \to -\infty$. Thus the saturated Bernoulli log likelihood is zero.

In practice this means $D = -2\ell(\hat{\pi})$ for ungrouped Bernoulli responses under the usual likelihood convention, but I would check grouping and binomial constants before using that shortcut in general.

Poisson and Gaussian deviance as comparison cases

For Poisson counts,

$$D = 2\sum_i \left[y_i \log \frac{y_i}{\hat{\mu}_i} - (y_i - \hat{\mu}_i) \right],$$

again with the appropriate limiting convention at $y_i = 0$. For Gaussian identity-link regression, the unscaled deviance equals the residual sum of squares, so deviance genuinely generalizes a familiar discrepancy measure.

MY SIDE NOTE

Information & IRLS

The IRLS weights arise from Fisher information, not an arbitrary regression trick.

Question I would ask: what assumption, derivative or geometric identity justifies this step?

A six-point logistic example

For $x = (0, 1, 2, 3, 4, 5)$ and $y = (0, 0, 1, 1, 0, 1)$, an illustrative logistic fit has coefficient estimates approximately $(-1.6841, 0.67365)$ and Bernoulli deviance about 6.77284. The identities $\sum \hat{\pi}_i = \sum y_i$ and $\sum x_i \hat{\pi}_i = \sum x_i y_i$ follow from the score equation for an intercept and one slope under a canonical logit link. These identities are practical checks; they do not hold without qualification for every link or model.

Based on: Deviance and three algorithms by hand and Chapter 5 GLM MLE. R checks: `glm(..., family=binomial())`, `logLik()`, `deviance()`, `predict(type="response")`.

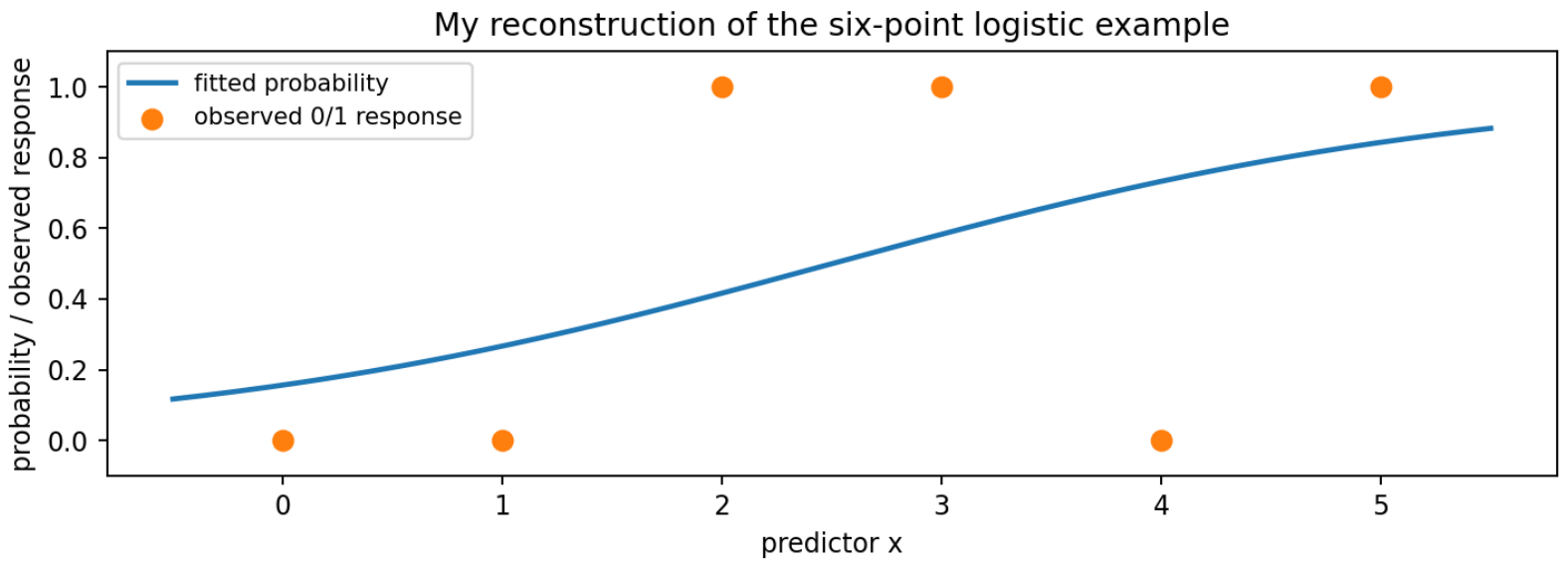


Figure 3: My reconstruction: six-point Bernoulli-logit curve and observed outcomes; this is an illustrative recreation using the handout's reported coefficients. R equivalents: `glm(family=binomial())`, `predict(type="response")`, base `plot()` and `lines()`.

VII. Family-by-family reference —the part I want beside my desk

Normal / identity

Natural parameter $\theta = \mu$, $V(\mu) = 1$, $g(\mu) = \mu$. The IRLS weight is constant and the coefficient solution reduces to ordinary least squares. I still need the variance scale for exact normal-model uncertainty statements.

Bernoulli / logit and binomial / logit

For Bernoulli, $\mu = \pi$, $\theta = \log[\pi/(1-\pi)]$, $V(\mu) = \mu(1-\mu)$, and the canonical link is logit. Binomial data can be written as successes out of m trials or as proportions with case weights; I must track which version my software uses. If $Y \sim \text{Bin}(m, \pi)$ is a count, $EY = m\pi$ and $\text{Var}\, Y = m\pi(1-\pi)$; the link convention should be stated in terms of π or μ to avoid mixing them.

Poisson / log

$\theta = \log \mu$, $V(\mu) = \mu$, $g(\mu) = \log \mu$, and $W_{ii} = \mu_i$. Log-link Poisson regression keeps all predicted means positive. If I model exposure, I can include an offset such as $\log t_i$; the resulting coefficient interpretation is a multiplicative rate ratio.

Exponential and Gamma / inverse or log

For an exponential observation with mean μ, the natural parameter is $-1/\mu$ and the variance function is μ^2. A Gamma model with fixed shape has the same $V(\mu) = \mu^2$, up to dispersion. The canonical link is inverse with an appropriate

MY SIDE NOTE

Deviance

I compare likelihood against a saturated model and handle boundary probabilities with limits.

Question I would ask: what assumption, derivative or geometric identity justifies this step?

sign convention, while the log link is a common practical alternative. With a log link, $g'(\mu) = 1/\mu$ and the IRLS weight is constant when $V(\mu) = \mu^2$.

Inverse Gaussian / inverse-square or log

For an inverse Gaussian mean parameterization, $V(\mu) = \mu^3$, and the canonical link is proportional to $-1/\mu^2$. A log-link version has $g'(\mu) = 1/\mu$, so $W_{ii} = 1/\mu_i$ (up to dispersion conventions). The mean domain restriction matters; I cannot let a numerical update generate an invalid prediction.

My compact table

Family	Mean domain	Variance function $V(\mu)$	Canonical link idea	Common alternative
Gaussian	real	1	identity	log for positive means
Bernoulli	$(0, 1)$	$\mu(1-\mu)$	logit	probit, cloglog
Poisson	positive	μ	log	identity with care
Gamma	positive	μ^2	inverse	log
Inverse Gaussian	positive	μ^3	inverse-square	log

The family controls the variance function; the link controls the path from μ to η. I choose them separately, then inspect deviance residuals and whether the assumptions are plausible.

Based on: GLM Family Reference. Useful R families/functions: `gaussian()`, `binomial()`, `poisson()`, `Gamma()`, `inverse.gaussian()`, `family()`, `glm.fit()`, `residuals(type="deviance")`.

VIII. Cross-validation —predicting on observations that were not used to fit

Three errors that I do not want to confuse

The training error is the average loss on the observations used to estimate the model. A test-set error uses separate held-out observations. The expected test error refers to the expected loss over future independent data and ordinarily cannot be observed directly. Cross-validation estimates predictive performance by repeatedly holding observations out.

The K-fold estimator

Partition the data into folds $I_1, \dots, I_K$. Let $\hat{f}^{(-k)}$ be a model fitted after excluding fold I_k. For a loss L,

$$\boxed{\widehat{\mathrm{Err}}_{\mathrm{CV}} = \frac{1}{n} \sum_{k=1}^{K} \sum_{i \in I_k} L\{y_i, \hat{f}^{(-k)}(x_i)\}}.$$

I prefer this formula to the vague instruction "run cross-validation": it tells me exactly what must be refitted, and on which data the loss is evaluated. If I select variables, impute missing values, standardize predictors, or tune hyper-parameters using the whole dataset before making folds, the validation data leak into the training procedure. Those operations belong inside each training fold.

LOOCV, K-fold, repeated CV and nested CV

Leave-one-out CV uses $K = n$ and refits leaving each observation out. It can be computationally expensive, although linear models have a shortcut. Ordinary five- or ten-fold CV balances the training fraction and computational cost. Repeated K-fold CV can reveal instability across partitions. Nested CV is important when I tune a model and also need an unbiased assessment of the entire tuning procedure: the inner folds choose settings, and the outer folds estimate held-out performance.

MY SIDE NOTE

Model families

I check V(mu) first, then link, scale, domain and weights.

Question I would ask: what assumption, derivative or geometric identity justifies this step?

The elegant LOOCV shortcut for linear regression

For OLS with a fixed model and full-rank design, the left-out prediction error can be written in terms of the ordinary residual and leverage:

$$e_i^{(-i)} = \frac{e_i}{1 - h_{ii}}.$$

Therefore

$$\text{LOOCV} = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{e_i}{1 - h_{ii}} \right)^2.$$

This equation is one of my favourite links in the course: the hat matrix from linear algebra reappears directly in out-of-sample prediction.

Choosing the loss to match the goal

For a continuous response I may use squared error or absolute error. For probabilities, log loss measures the quality of probabilistic predictions; for classification, misclassification error depends on the threshold and ignores probability calibration. For count models, Poisson deviance can be more natural than raw squared error. I should write down the loss before fitting instead of choosing it after seeing which model performs best.

When random folds are inappropriate

Observations from the same patient, household, hospital or time series may not be independent. Random folds can then leak information and give an overly optimistic estimate. Grouped CV, blocked CV, or rolling-origin validation may match the actual prediction setting better. The validation scheme is part of the scientific question, not just a software option.

Based on: MTH303 Cross-Validation handout. R equivalents: `sample()`, `split()`, `predict()`, `glm()`, `lm()`, `hatvalues()`, `boot::cv.glm()`, or `rsample::vfold_cv()`.

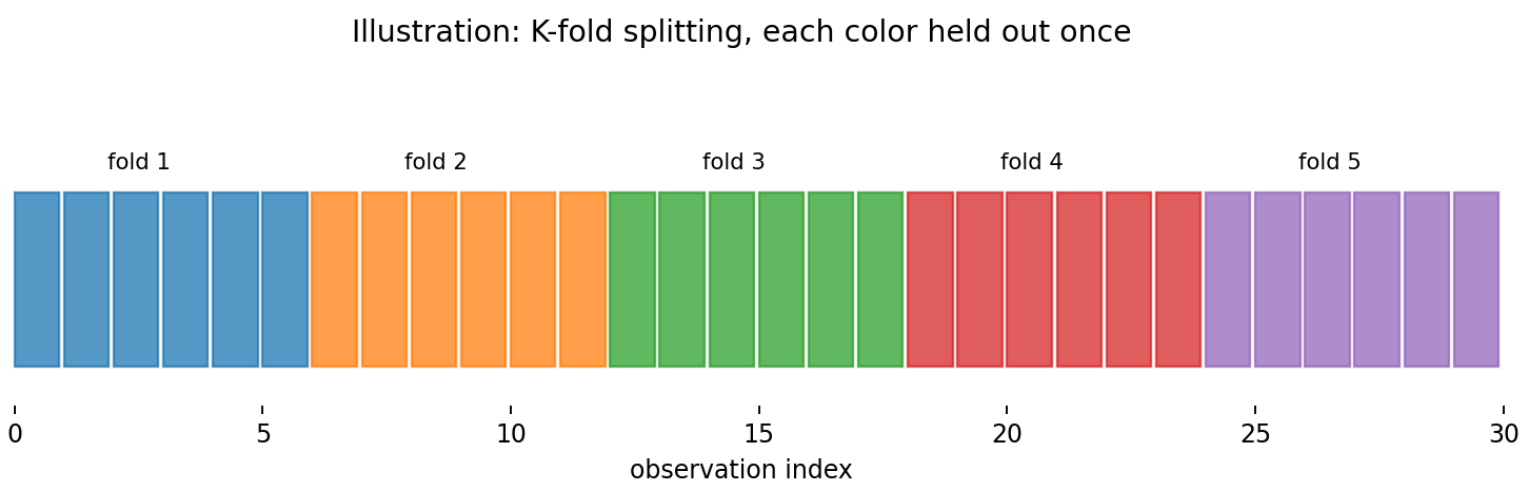


Figure 4: My reconstruction: illustrative fold assignments and the held-out predictions idea. R equivalents: `sample()`, `split()`, `rsample::vfold_cv()`, `ggplot2::geom_tile()`.

IX. My proof-checking list

1. **Before invoking a distribution:** which assumptions give the result? Projection geometry needs fewer assumptions than exact t/F sampling distributions.
2. **Before differentiating GLM likelihood:** is the parameter θ, μ, η or β? Write every derivative in the chain.

MY SIDE NOTE

Cross-validation

Validation requires held-out predictions and refitting preprocessing within each training fold.

Question I would ask: what assumption, derivative or geometric identity justifies this step?

3. **Before calling a link canonical:** does it identify η with the natural parameter under the actual exponential-family parameterization?
4. **Before doing IRLS:** how are W and the working response obtained? Do they agree with the score and Fisher information?
5. **Before using a deviance:** which saturated model, constants, dispersion conventions and zero-count limits are involved?
6. **Before comparing AIC or BIC:** are the likelihoods comparable and the fitted datasets identical?
7. **Before interpreting CV:** were preprocessing, feature selection and tuning performed inside the training folds?
8. **Before interpreting numerical output:** do rank, separation, extreme weights, influence and the response domain undermine the computation?

The broader idea I want to keep is that **geometry, probability and computation are three views of one modeling process**. I learned the projection picture in linear algebra; I understand likelihood through mathematical statistics; and I now see why numerical analysis matters when I actually fit the model.

Sources I used for these additional notes

These are the eight supplementary resources linked from the MTH303 teaching page. I did **not** reproduce the entire source pages or their full code listings; the explanations, derivations and illustrative figures above are newly written course notes.

1. Chapter 3: Multiple Linear Regression
2. Model Selection Criteria
3. Chapter 4: GLM Proofs
4. Canonical and Non-canonical Links
5. Chapter 5: GLM MLE
6. GLM Family Reference
7. Deviance and Three Algorithms by Hand
8. Cross-Validation

MY SIDE NOTE

Last connections

Geometry, inference and numerical algorithms are describing the same fitting process.

Question I would ask: what assumption, derivative or geometric identity justifies this step?